



How earthworms and plants inspire scientists creating robots

We enjoyed the return of the Pudding Seminar series on Friday with a fascinating exploration of how robots are designed. Chapa Sirithunge, Postdoctoral Affiliate, revealed how careful analysis of the mechanics of movement in the natural world can inspire their development.

In exploring the world, embodied robots demonstrate that a physical body is as crucial as intelligence. While traditional robotics has focused primarily on computational intelligence, recent advances highlight the importance of physical embodiment. A robot's body influences its interactions with the environment, enabling it to perform tasks and navigate spaces in ways purely computational systems cannot.

There are many natural sources scientists draw on to create robots. This 'bio-inspiration' includes the setal arrays on the feet of geckos, a dense arrangement of microscopic hair-like structures which help them to climb vertically. A range of other plants and creatures helped to inspire different robotic movements including:

- The **decentralised control** of octopus and human eyes that can help create grabbing objects which sense and adapt to different shapes.
- The **adaptive shape** of pinecones that open in response to a changing environment, which again can help create robotic hands that change shape to enable the lifting of

- And the **adaptive mechanics** of a tendril serving as a spring-damper system to support a plant.

In her talk, *Navigating the Physical World: The Essential Role of Robots*, Chapa also showed robotic hands at different stages of development, handing them round along to show the hard and soft materials. These included a robotic hands made of silicone over hard plastics to mimic the skeletal underlay and soft encasing of a human hand; and another wired for movement.

Chapa completed a PhD in Electrical Engineering at University of Moratuwa, Sri Lanka before joining the University of Cambridge as a postdoctoral research associate in 2022 and currently works as a Marie-Curie fellow, as well as being a postdoctoral affiliate at Newnham College. She recently won an award through the Postdoc Academy as a **Heroine of Science** for her inspiring journey as a female scientist.

In highlighting the importance of a robot's body, Chapa drew an analogy with growing understanding of the strong coupling of body and mind: for instance, the stomach affects the human brain as if we are hungry we can behave differently.

Chapa is part of the Women in Robotics network –a support network established by Cambridge roboticists to help women get in to the field and thrive. The network is in its initial stages and there will be a set of events scheduled for this term and next year. You can join this network and receive updates by completing [this form](#).

Details of all our Pudding Seminars and their format can be found [via the Research section of this website](#).



Chapa Sirithunge
Postdoctoral Affiliate

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